Wyniki etapu II: Specyfikacja i analiza wymagań na oprogramowanie

<EcoRoute>

Projektowanie oprogramowania

Skład zespołu:
Michał Strugarek
Alina Lenart
Błażej Kowal
Marek Gagaczowski

Prowadzący: dr inż Marcin Jodłowiec

Specyfikacja wymagań

Guest

As a Guest, I want to be able to register in the system when I don't have an account.

As a Guest, I want to be able to log in to the system.

User

As a User, I want to have access to my Route History so that I can easily review and compare pollution profiles from my previous activities.

As a User, I want the system to automatically and periodically record pollution data with coordinates during an active route to ensure high resolution and accuracy of my route tracking.

As a User, I want full control over my personal data and the ability to configure GDPR consent for storing and processing my location data in Account Settings, so that I can be confident my data is protected according to regulations.

As a User, I want the system to automatically generate a detailed Route Report after each activity, including pollution data, so that I can analyze the total impact of pollution on my health during that activity.

As a User, I want to be able to Report Inaccurate Sensor so that I can notify the system when pollution readings seem incorrect, enabling the platform to flag potential sensor issues, improve data quality, and ensure that the information used in my reports and analyses is reliable.

As a User who may forget to stop recording, I want the system to automatically end the route after a certain time limit to avoid battery drain and ensure the accuracy of collected data.

As a User, I want to be able to change my password to enhance the security of my account and personal data.

Sensitive User

As a Sensitive User (e.g., an asthma sufferer), I want to receive immediate, critical notifications when the current pollution level exceeds the defined Health Limit so that I can minimize the risk of severe health deterioration.

Admin

As an Admin, I want to be able to add, edit, and delete measurement stations to maintain the accuracy and timeliness of air pollution data.

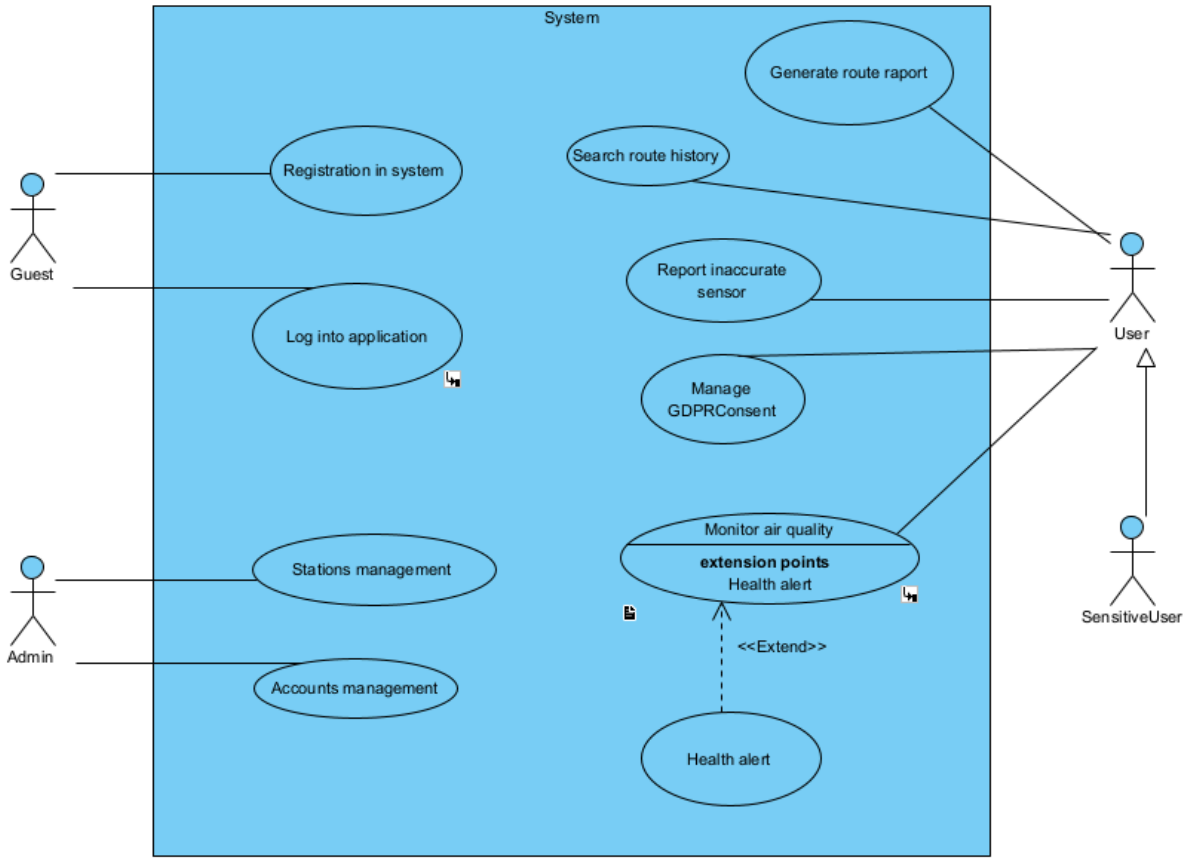
As an Admin, I want to define and adjust pollution thresholds according to the latest WHO guidelines so that the system can accurately generate and send appropriate alerts to users.

As an Admin, I want to have access to general system usage statistics, including the number of active users and recorded routes, so that I can monitor system performance and plan its development.

As an Admin, I want to view, block, unblock, and reset user accounts and passwords to effectively manage system security and assist users with access issues.

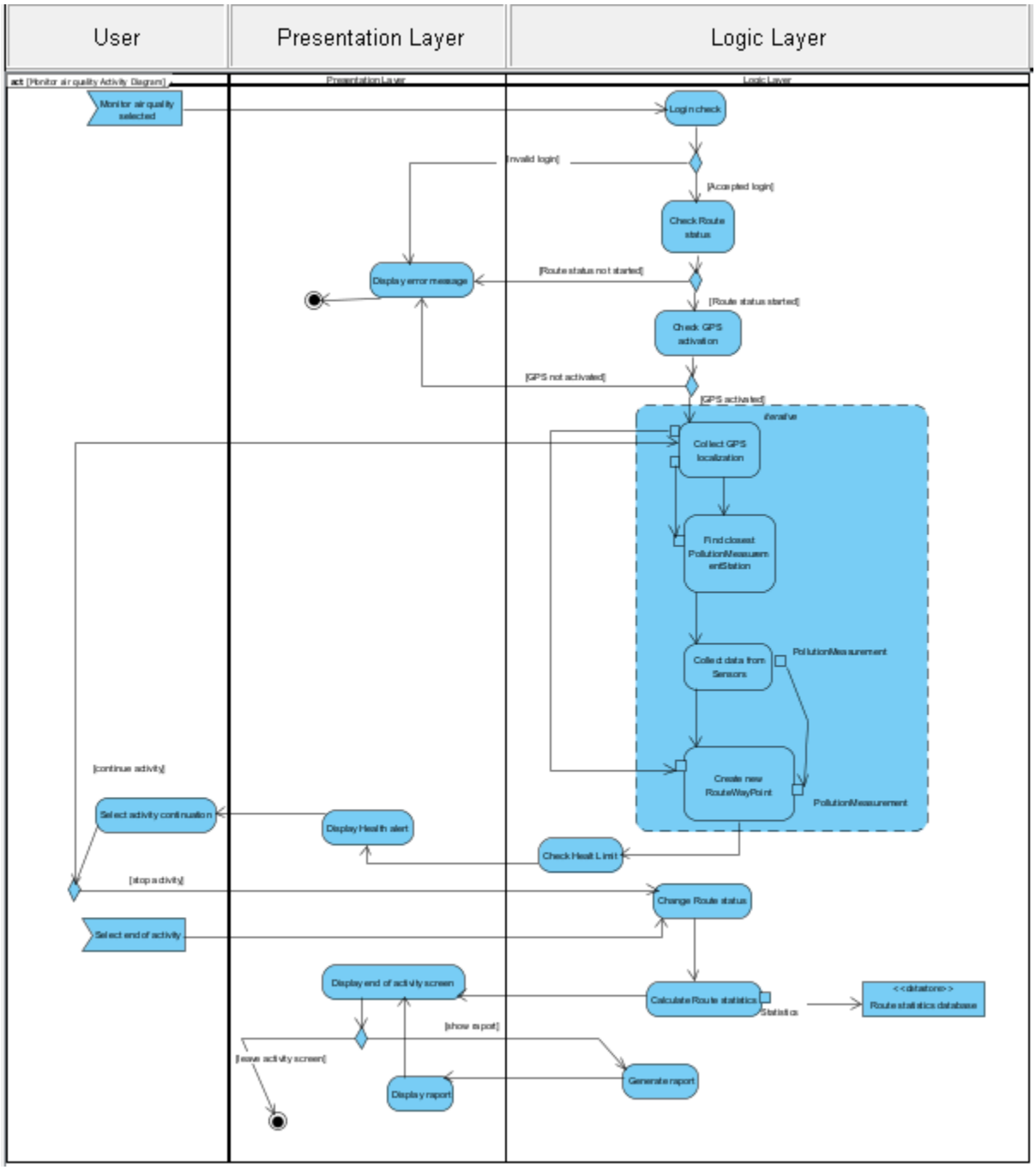
Model przypadków użycia i prototypy interfejsów

1. Diagram przypadków użycia

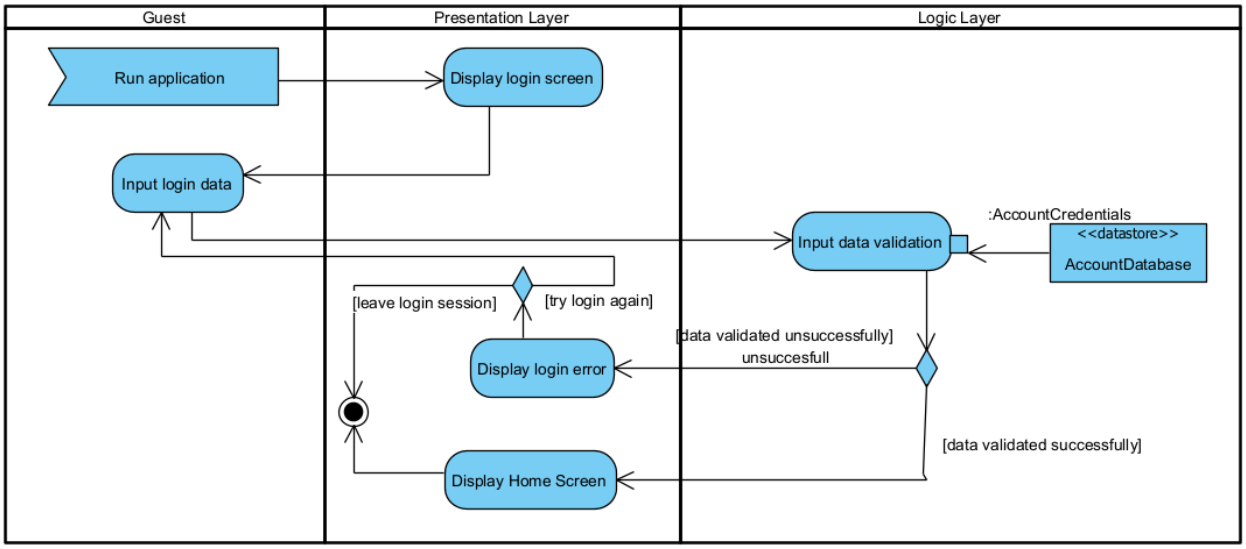


2. Specyfikacja przypadków użycia i Diagramy aktywności

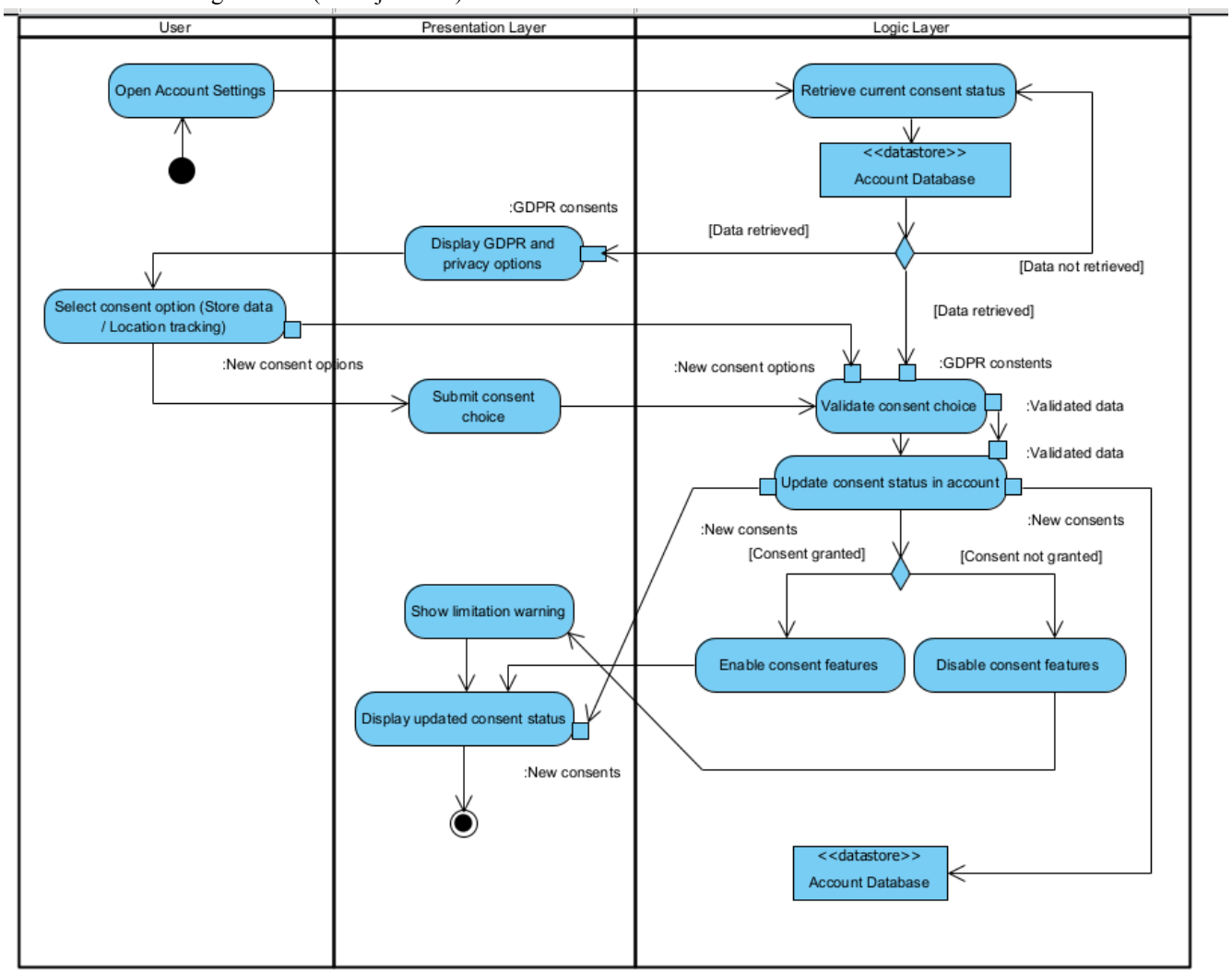
2.1.1 User - Monitor air quality (Marek Gagaczowski)



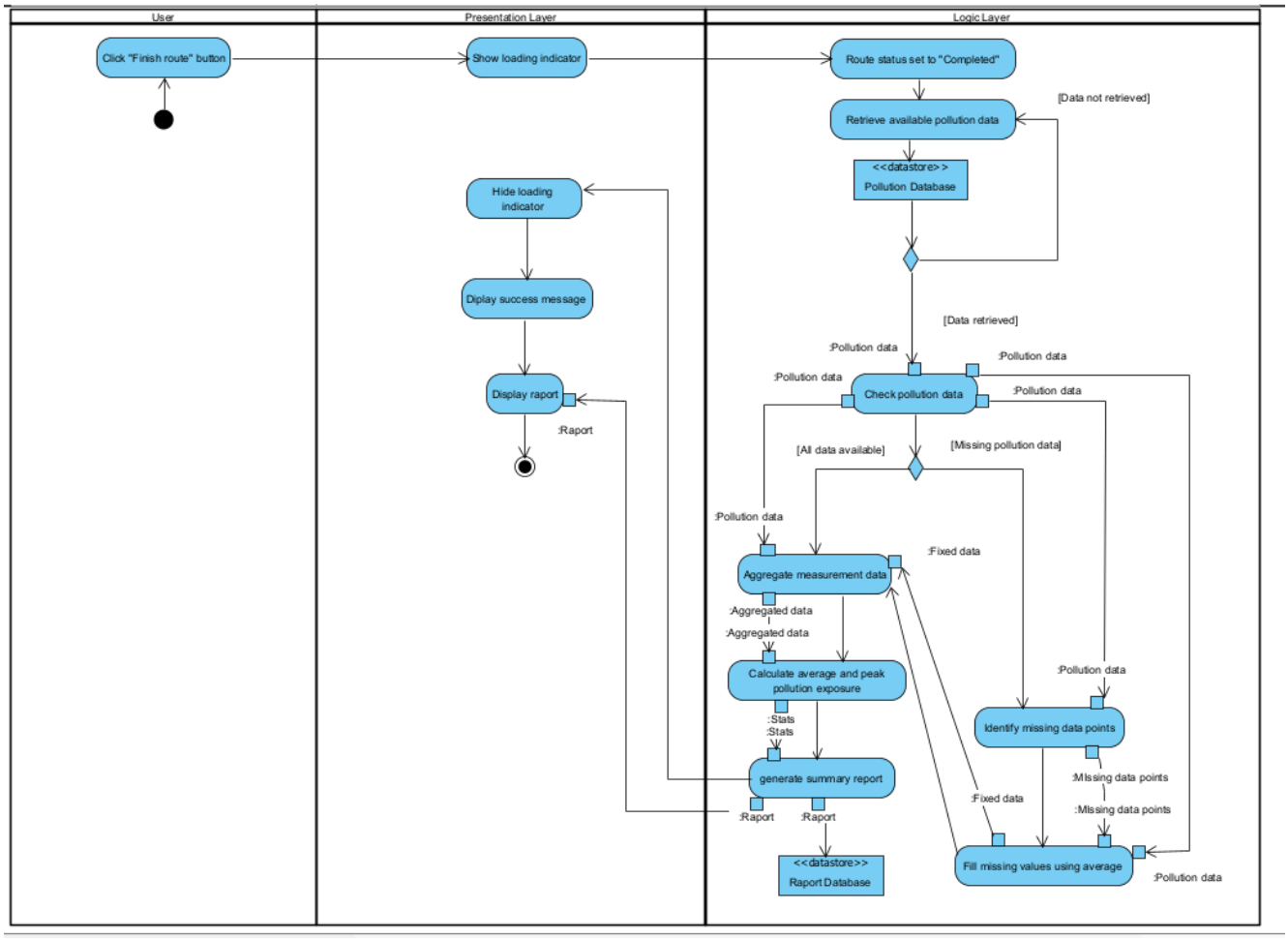
2.1.2 Guest - Log into application (Marek Gagaczowski)



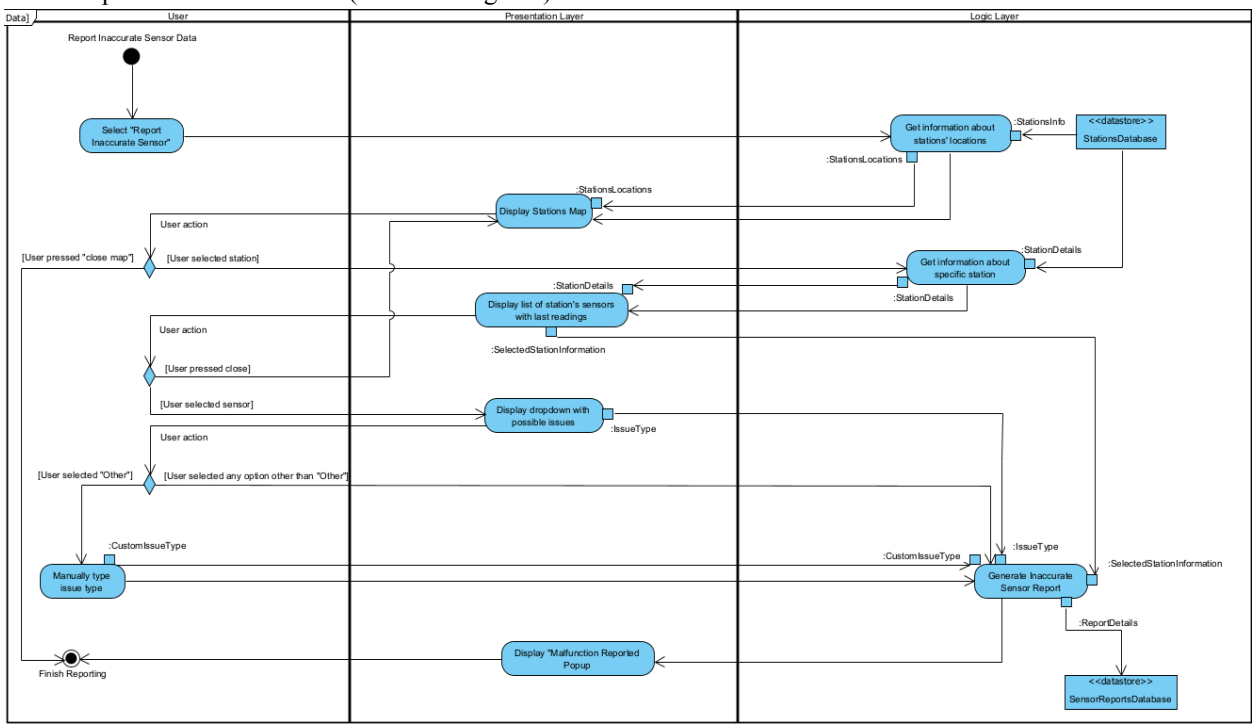
2.1.3 User - Manage GDPR (Błażej Kowal)



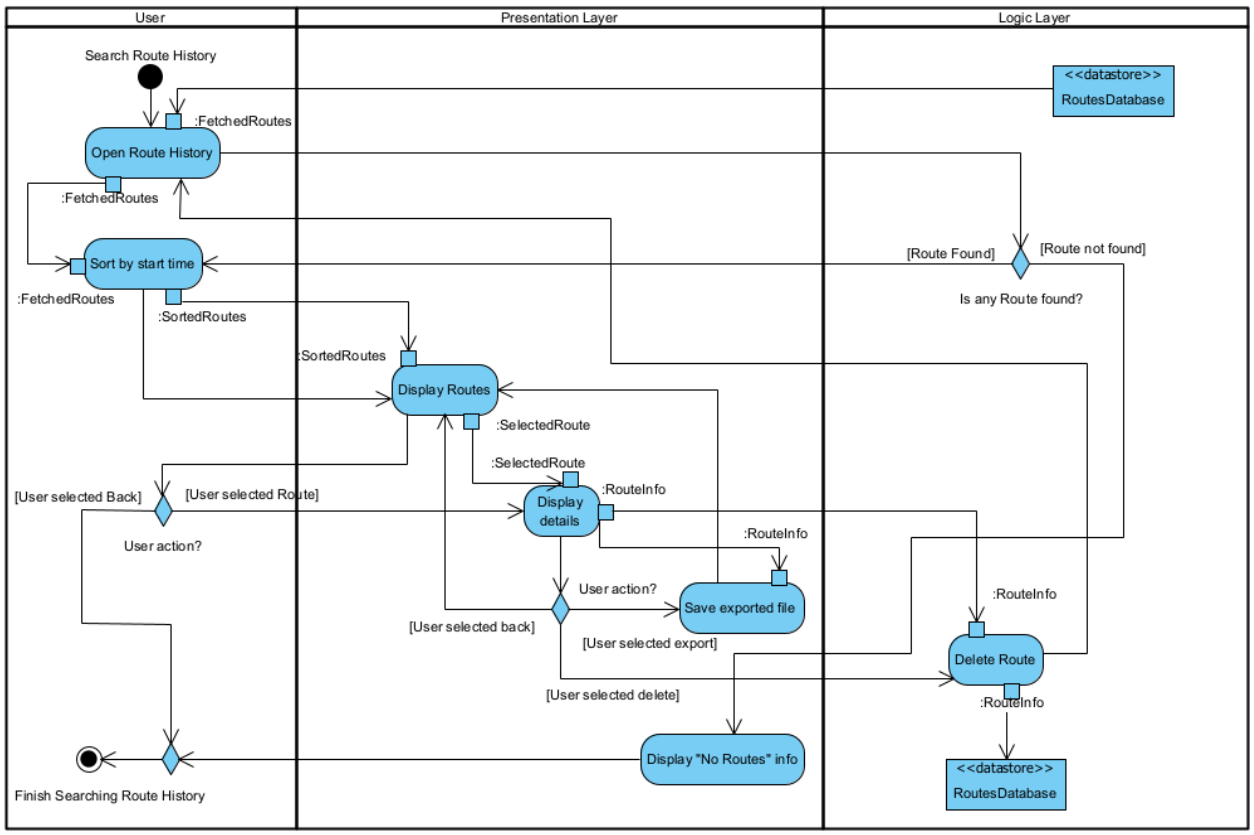
2.1.4 User - Generate route raport (Błażej Kowal)



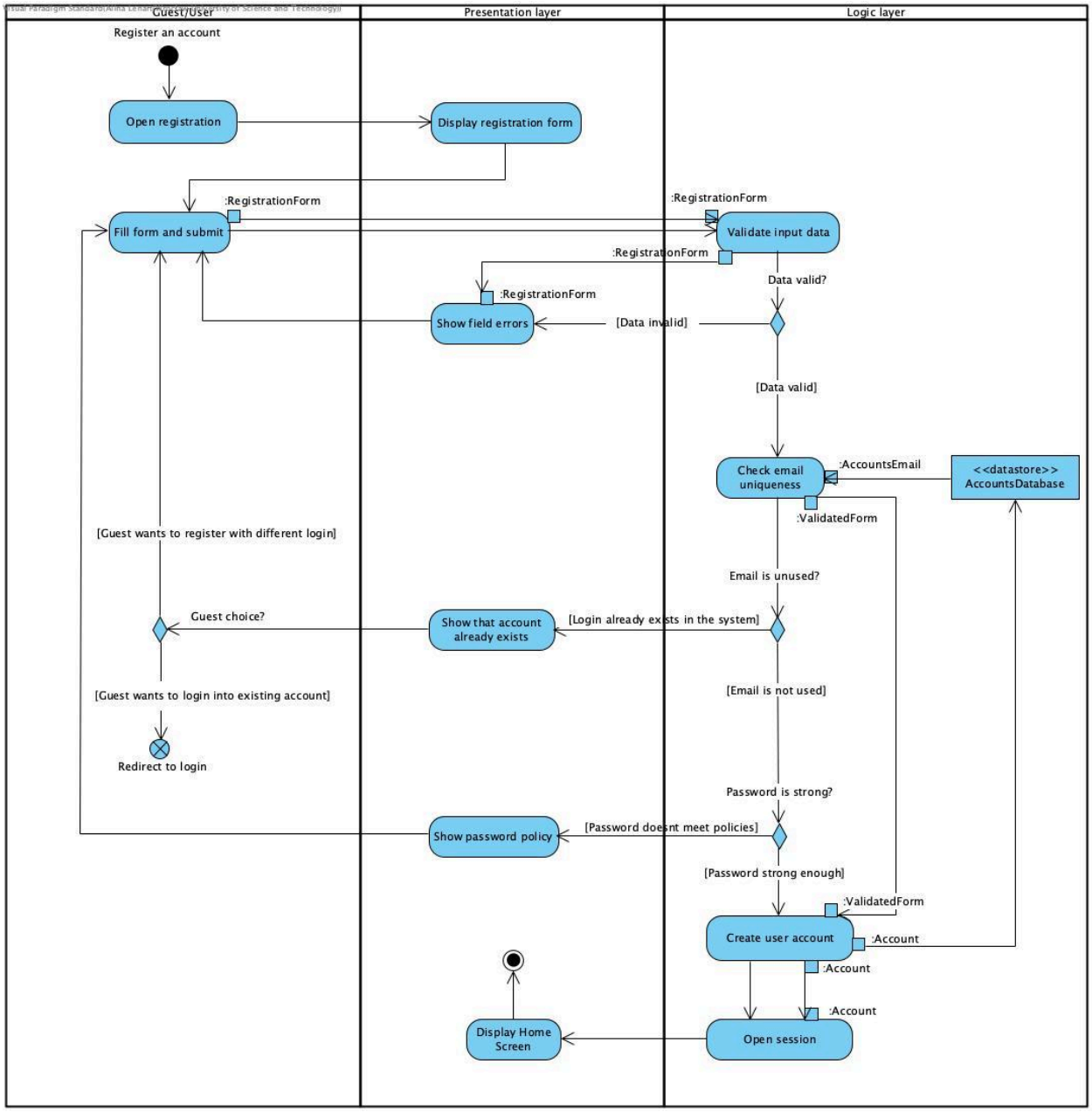
User - Report Inaccurate Sensor (Michał Strugarek)



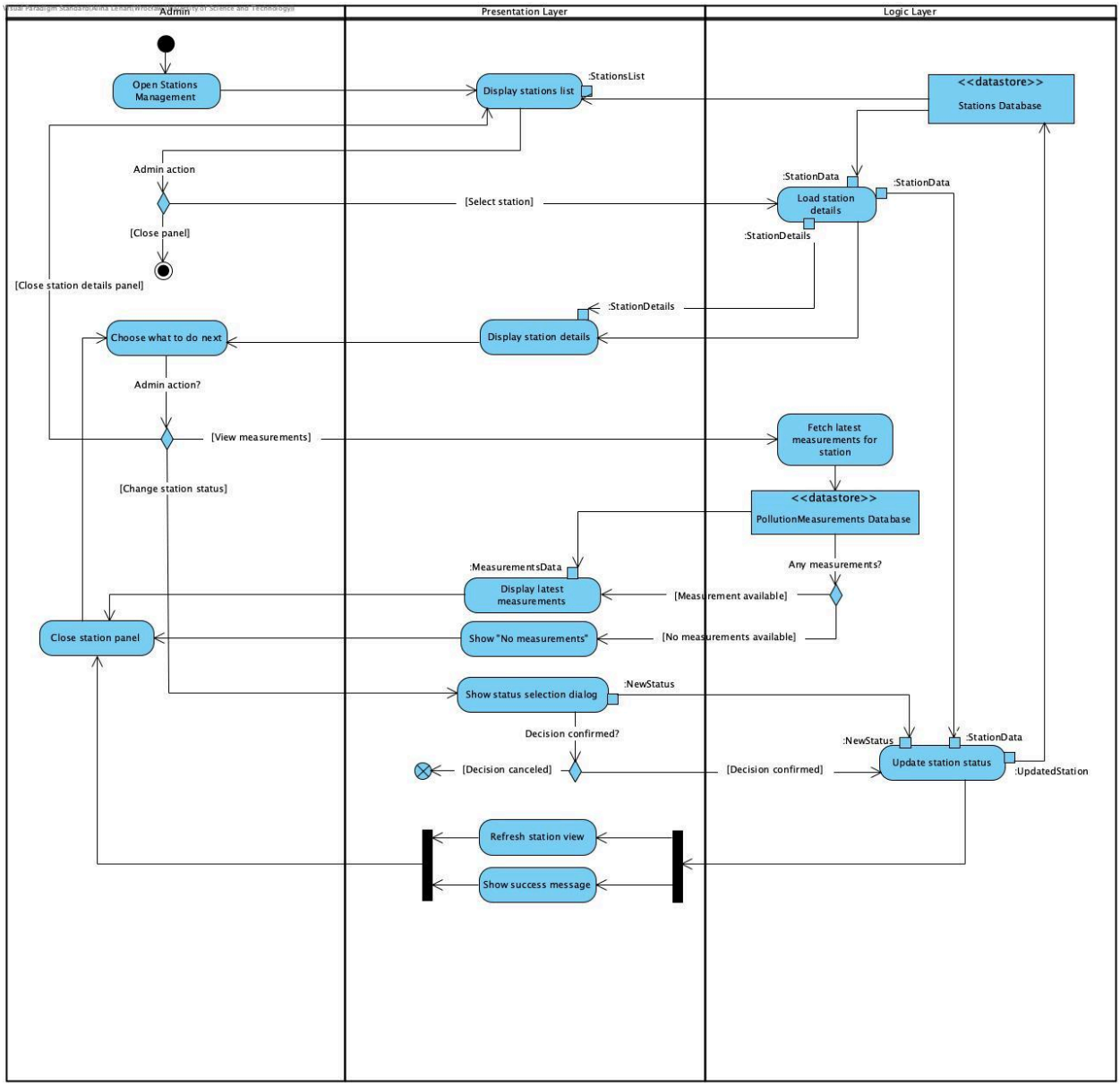
2.1.5 User - Search Route History (Michał Strugarek)



2.1.6 Guest - Registration In System (Alina Lenart)



2.1.7 Admin - Stations Management (Alina Lenart)



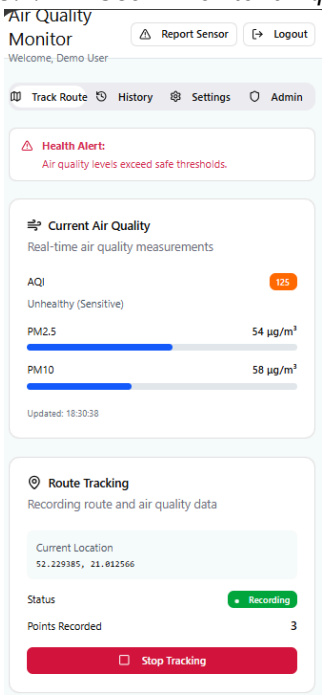
3. Prototyp interfejsu

Style Guide - Air Quality Monitor

Primary colors: Primary #030213 (dark navy), Background #FFFFFF, Destructive #D4183D (red), Muted #ECECF0 (light gray)
Semantic colors: Success green, Warning yellow/orange, Error red, Links/Actions blue, Admin purple
Typography: System sans-serif font stack, monospace for technical data; base size 16px, line-height 1.5
Font sizes: 12px (xs), 14px (sm), 16px (base), 18px (lg), 20px (xl), 24px (2xl)
Font weights: 400 normal (body), 500 medium (labels, buttons), 600 semibold (emphasis)

Headings: H1=24px, H2=20px, H3=18px, H4=16px - all medium weight
Border radius: 6px (sm), 8px (md), 10px (lg/default), 14px (xl), full/pill for badges
Buttons: Primary - dark background, white text, 8-16px padding, 8px radius; Outline - 1px border, transparent bg; Destructive - red background
Inputs: Light gray background #F3F3F5, 8px radius, 8-12px padding, focus ring on interaction
Cards: White background, 1px subtle border, 10px radius, 24px padding
Spacing: Base unit 4px; sections 24px apart, elements 8-16px gap, container padding 16-32px
Transitions: 150ms duration, smooth easing cubic-bezier(0.4, 0, 0.2, 1)

3.1.1 *User - Monitor air quality (Marek Gagaczowski)*



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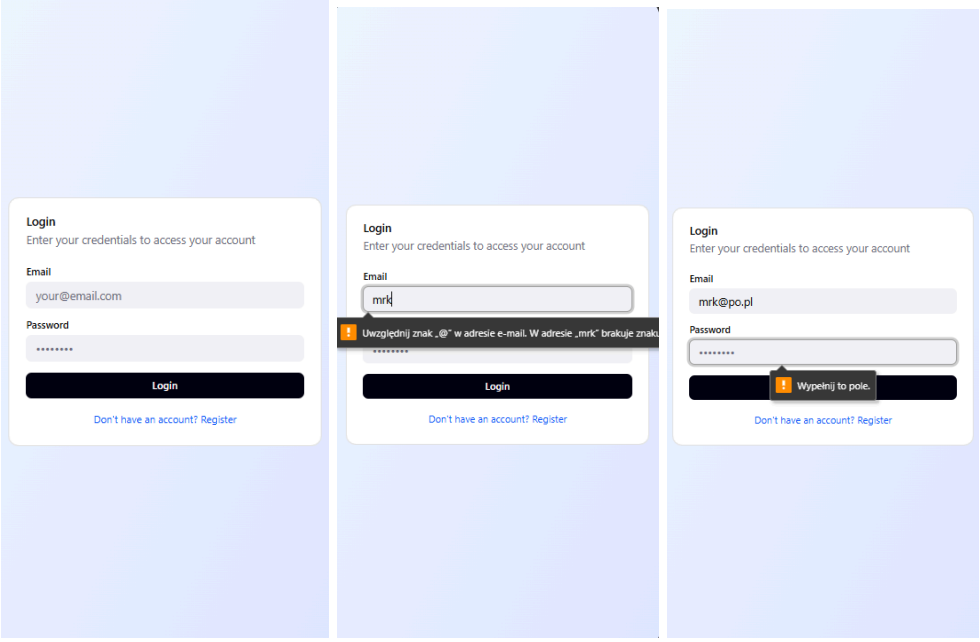

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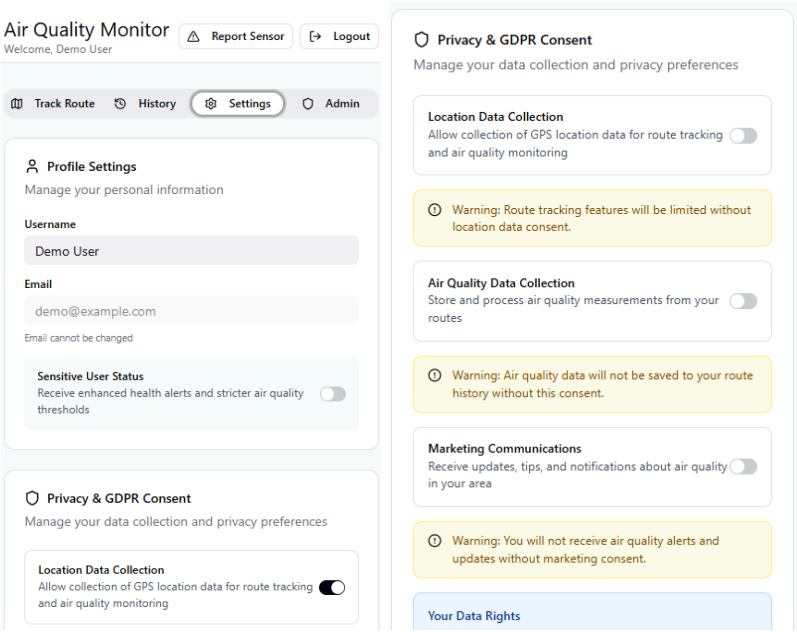

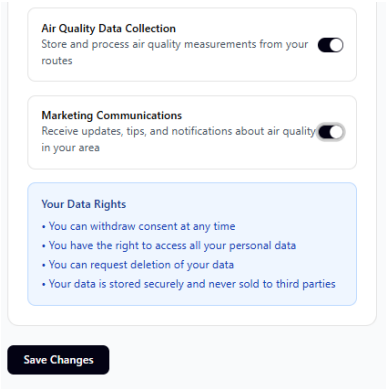

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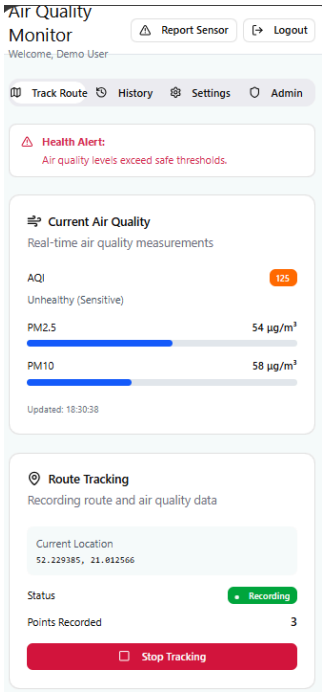

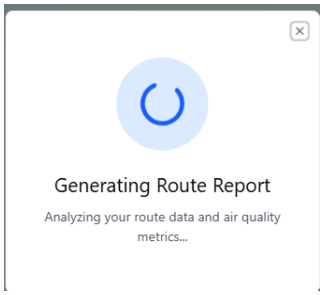

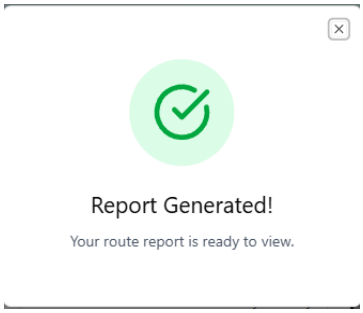

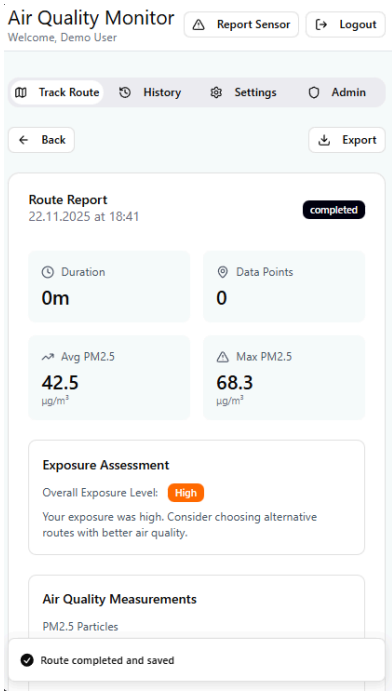

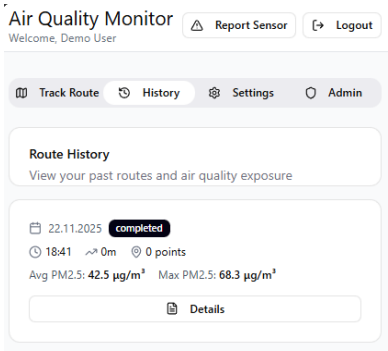

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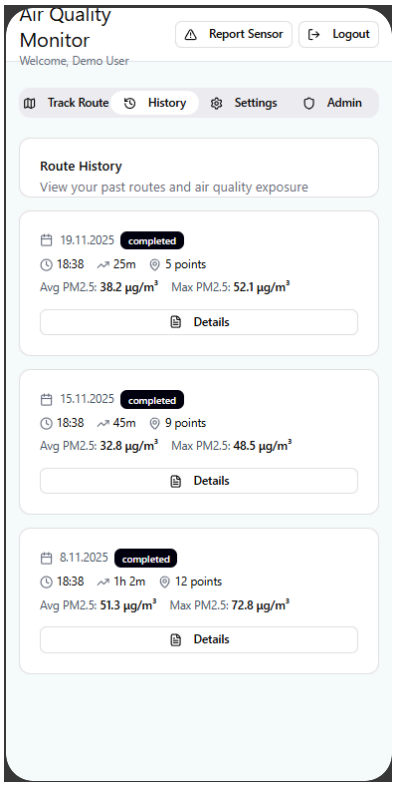

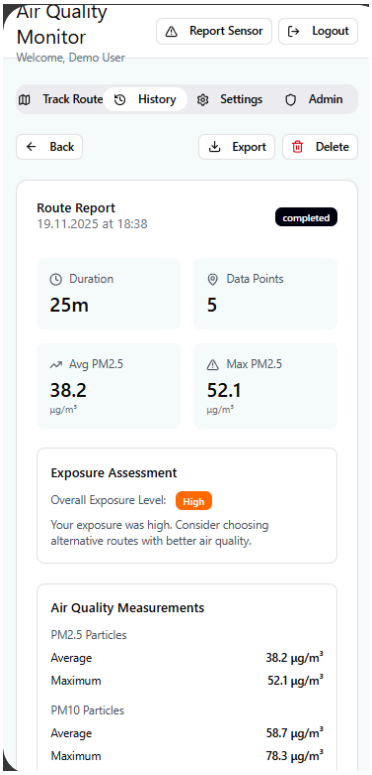

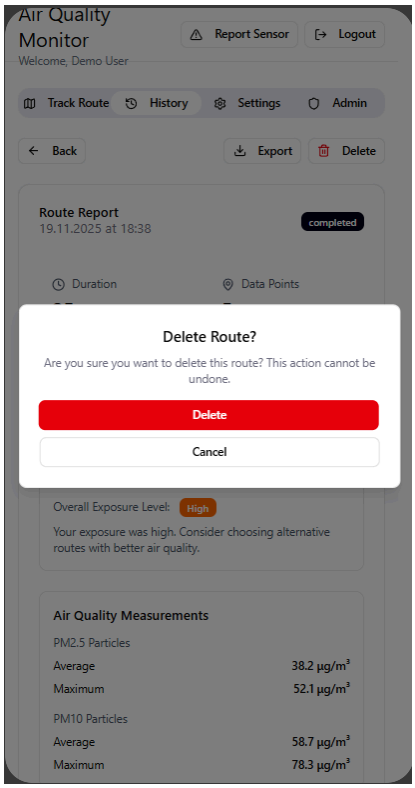

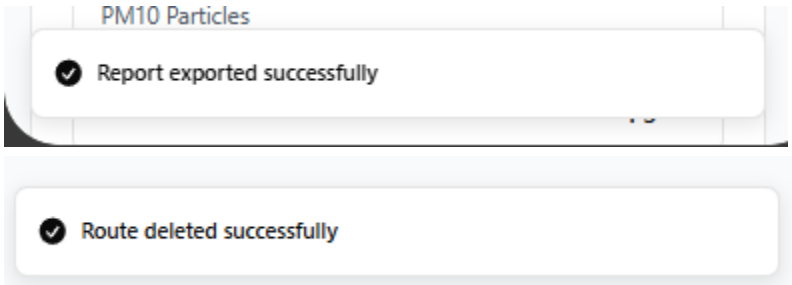

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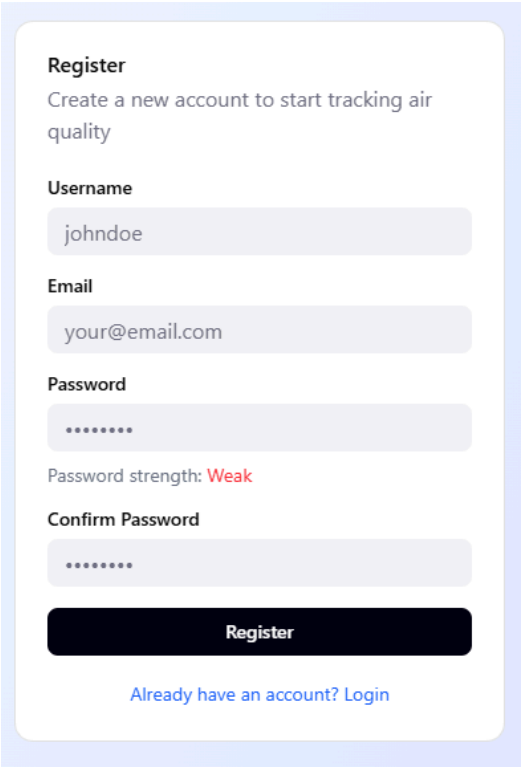

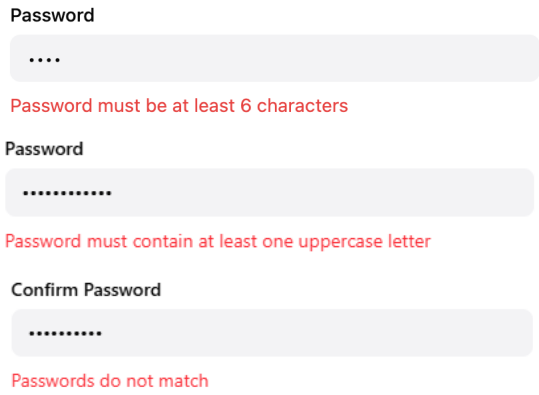

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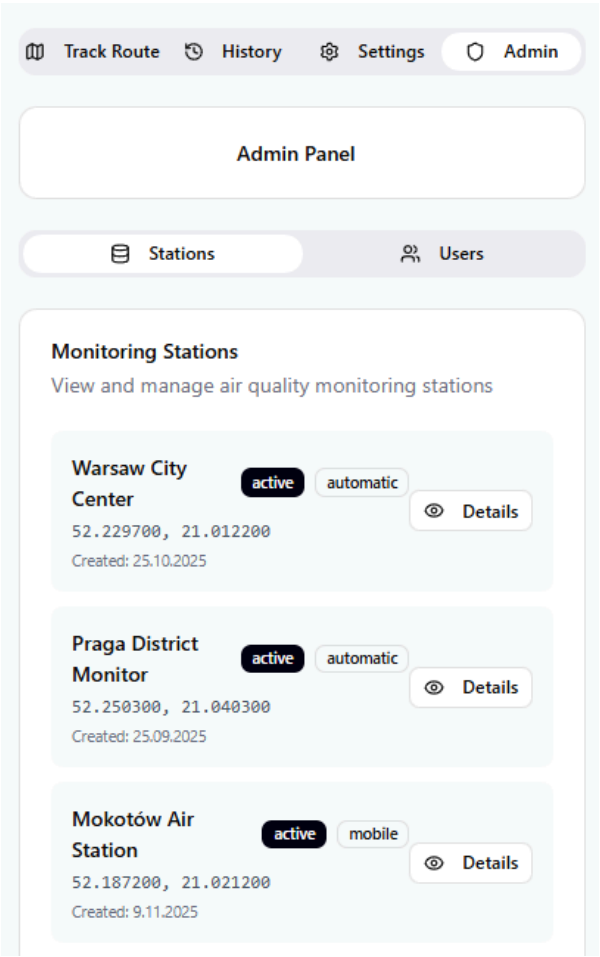

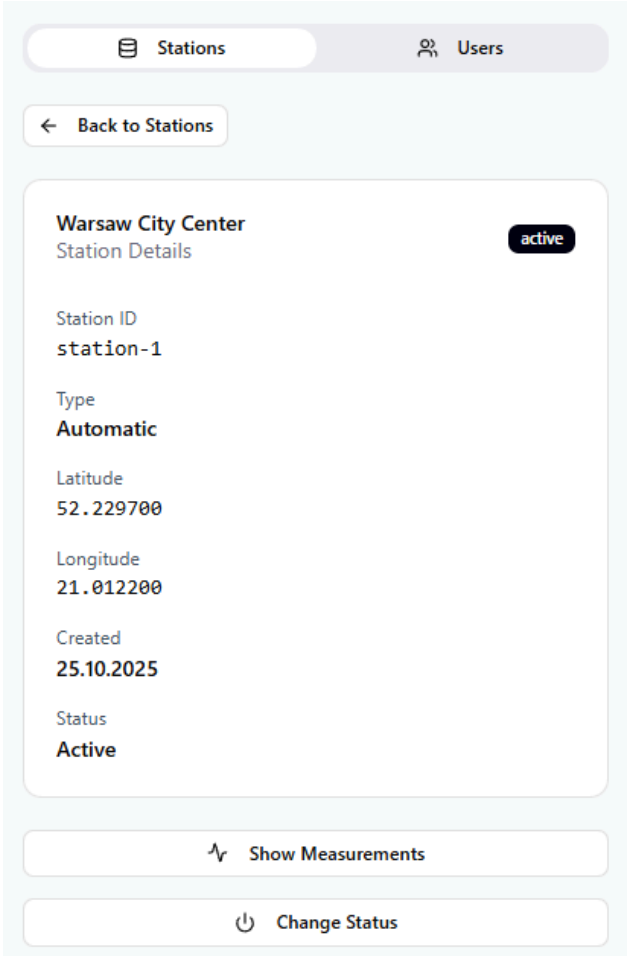

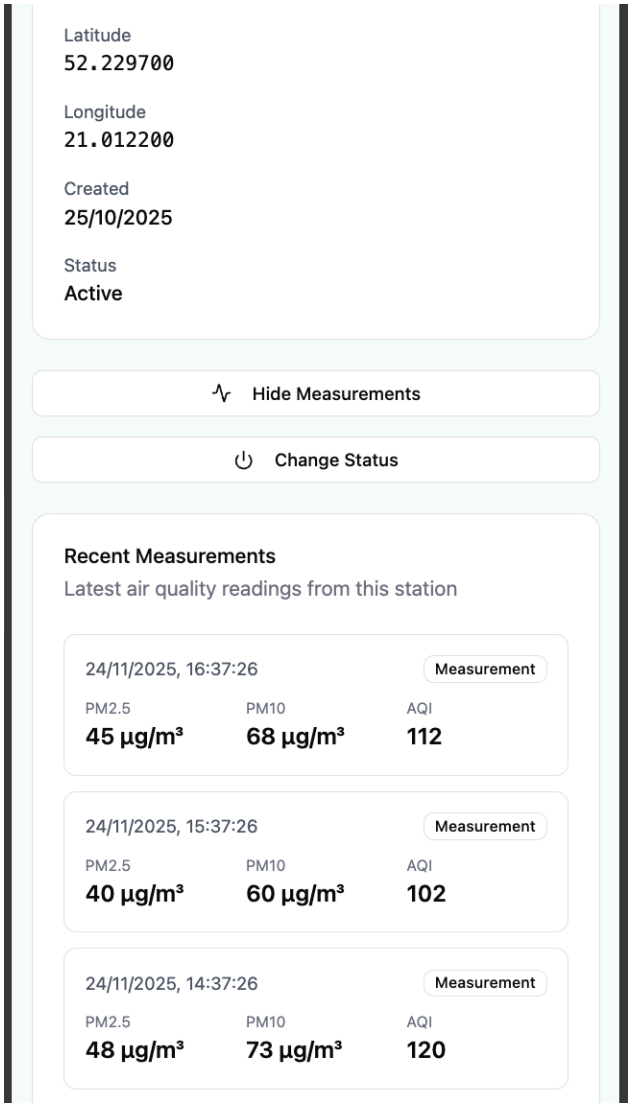
Latitude
52.229700
Longitude
21.012200
Created
25/10/2025
Status
Active
Hide Measurements
Change Status
Recent Measurements
Latest air quality readings from this station
24/11/2025, 16:37:26
Measurement
PM2.5
45 µg/m³
PM10
68 µg/m³
AQI
112
24/11/2025, 15:37:26
Measurement
PM2.5
40 µg/m³
PM10
60 µg/m³
AQI
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24/11/2025, 14:37:26
Measurement
PM2.5
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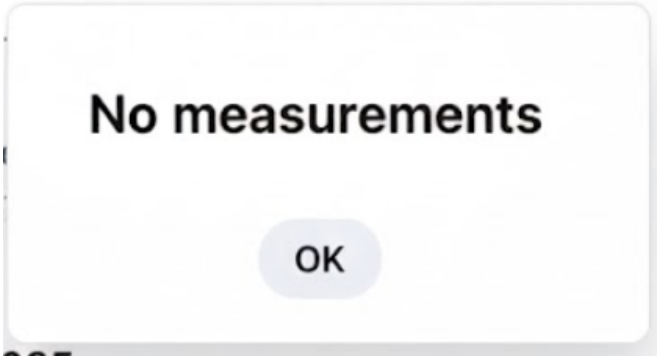
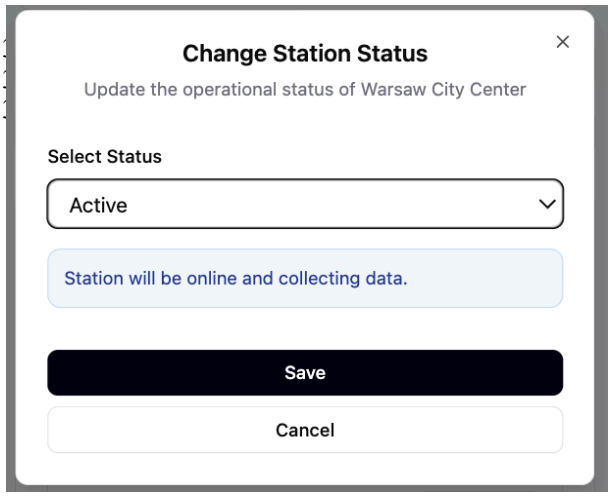
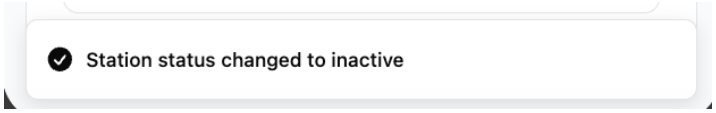
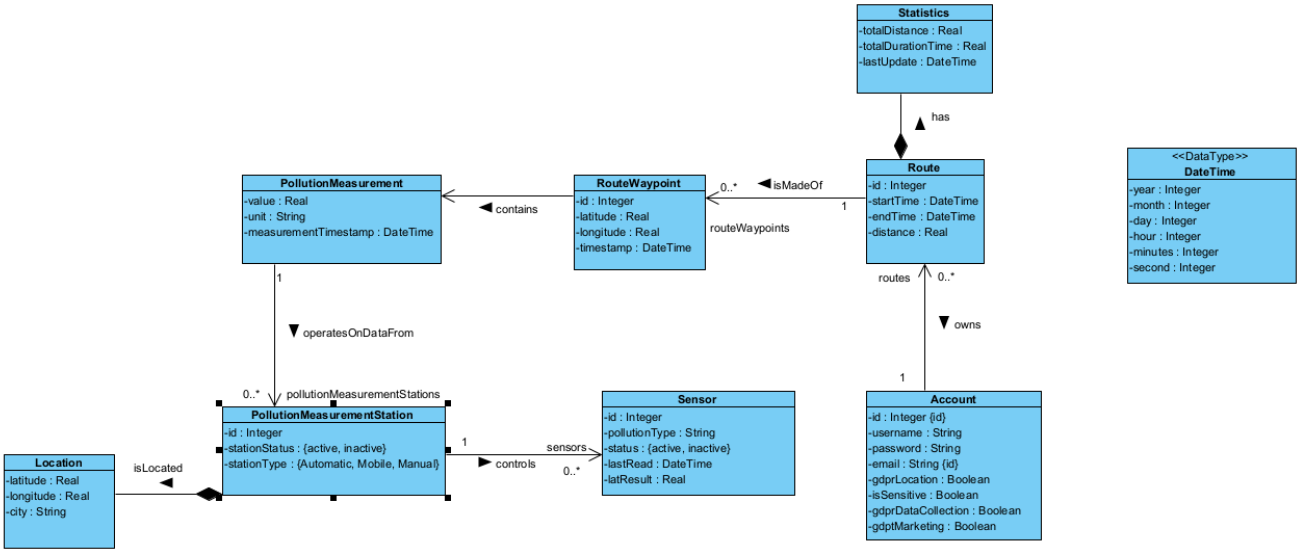
PM10
73 µg/m³
AQI
120


No measurements
OK


Change Station Status
Update the operational status of Warsaw City Center
Select Status
Active
Station will be online and collecting data.
Save
Cancel


Station status changed to inactive

Model informacyjny

[Class diagram with additional comments when necessary.]